Mohamad Chehade

Ph.D. Student · Electrical and Computer Engineering

The University of Texas at Austin

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Education

The University of Texas at Austin *Austin, Texas*

Ph.D. in Electrical and Computer Engineering *Aug. 2023 – Present*

- Advisor: Dr. Hao Zhu
- Research Interests: Large Language Model Alignment, Time Series Foundation Models, Risk-aware Transfer Learning, Learning for Optimization and Control, Neural Network Verification and Certified Robustness

The University of Texas at Austin *Austin, Texas*

Master of Science in Engineering *Aug. 2023 – May 2026*

- Advisor: Dr. Hao Zhu
- Relevant Courses: Reinforcement Learning, Generative AI, Statistical Machine Learning, Applied Machine Learning, Probability and Stochastic Processes, Applied Stochastic Processes, Learning-based Optimal Control, Convex Optimization, Energy Optimization and Operation

American University of Beirut *Beirut, Lebanon*

B.Eng. in Electrical and Computer Engineering *Aug. 2019 – Jun. 2023*

- GPA: 4.0/4.0 | Minor in Mathematics
- Final Year Project: Optimal Power Flow via Machine Learning (Advisor: Dr. Rabih Jabr)
- Undergraduate Research: Microgrid Sizing using Ordinal Optimization (Advisor: Dr. Sami Karaki)

Experience

Argonne National Laboratory - Grid Optimization and Infrastructure Group *Remote*

Research Aide – Mentors: Dr. Feng Qiu & Dr. Wei Gao *May 2026 – Aug. 2026*

- Developing decision-aware time series foundation models that align forecasting with downstream decision-making and optimization.

Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group *Remote*

Graduate Research Assistant (GRA) - Mentor: Dr. Russell Bent *May 2025 - Aug. 2025*

- Extended neural network verification ideas toward robustness training and certified robustness
- Developed a training algorithm that leverages verifiable input regions to improve certified robustness of neural networks

Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group *Los Alamos, NM*

Graduate Research Assistant (GRA) - Mentors: Dr. Wenting Li, Dr. Brian Bell *Jun. 2024 - Aug. 2024*

- Worked on the verification of neural networks in physical and safety-critical systems
- Developed two algorithms for determining large verifiable input regions of neural networks

University of Connecticut - Center for Clean Energy Engineering - PEARL Lab *Storrs, CT*

Research Internship - Advisor: Dr. Ali Bazzi *Jun. 2022 - Aug. 2022*

- Collaborated with Ward Leonard, a leading industrial motor manufacturer, on fault-tolerant power electronics
- Developed a fault diagnosis algorithm for power electronic inverters using combinational logic
- Optimized and constructed the inverter circuit for high-power testing using mixed-integer linear programming

Publications

Published / Accepted

Chehade, M., & Zhu, H. (2026). NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS 2026)*. arXiv:2509.21668 | Code

Chehade, M., & Zhu, H. (2026). Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. Accepted to *2026 IEEE Power & Energy Society General Meeting (PESGM 2026)*.

Cho, Y.-H., **Chehade, M.**, Al-Janahi, F., Lim, S., Mohammadi, J., & Zhu, H. (2026). Carbon-aware Market Participation for Building Energy Management Systems. Accepted to *2026 IEEE Power & Energy Society General Meeting (PESGM 2026)*.

Chehade, M., Li, W., Bell, B. W., Bent, R., Kazi, S. R., & Zhu, H. (2025). LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2408.08824 | OpenReview

Chehade, M., Ghosal, S. S., Chakraborty, S., Reddy, A., Manocha, D., Bedi, A. S., & Zhu, H. (2025). Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2505.23729 | OpenReview

Chehade, M., & Karaki, S. (2025). BOOST: Microgrid Sizing Using Ordinal Optimization. *2025 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, pp. 1–4. doi:10.1109/TPEC63981.2025.10907217.

Chehade, M., Cho, Y.-H., Chinchali, S., & Zhu, H. (2024). Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *2024 56th North American Power Symposium (NAPS)*, El Paso, TX, USA, pp. 1–5. doi:10.1109/NAPS61145.2024.10741791.

Under Review

Chehade, M., Bedi, A. S., Chakraborty, S., Zhang, A., & Zhu, H. (2026). Test-Time Risk Adaptation with Mixture of Agents. Under review at *NeurIPS 2026*. arXiv:2408.08812

Skills

- **Programming Languages:** Python, MATLAB, C/C++, SQL, R
- **ML / Scientific Computing:** PyTorch, NumPy, SciPy, Scikit-learn, Matplotlib, Gurobi, MATPOWER
- **Engineering & Power Systems Tools:** Simulink, SPICE, HOMER, PVSyst, LabVIEW
- **Tools & Workflow:** Git, LaTeX, Linux, Overleaf, Microsoft Office Suite
- **Languages:** English (fluent), French (fluent), Arabic (native)

Reviewer

Sep. 2023 - Present	**IEEE Transactions on Smart Grid**
July 2024 - Present	**Asilomar Conference on Signals, Systems, and Computers**
Dec. 2024 - Present	**Texas Power and Energy Conference (TPEC)**
June 2025 - Present	**Hawaii International Conference on System Sciences (HICSS)**
Sep. 2025 - Present	**IEEE Systems Journal**
Oct. 2025 - Present	**International Conference on Learning Representations (ICLR)**
Feb. 2026 - Present	**International Conference on Machine Learning (ICML)**
May 2026 - Present	**IEEE Power Engineering Letters**
June 2026 - Present	**Conference on Neural Information Processing Systems (NeurIPS)**
Aug. 2026 - Present	**AAAI Conference on Artificial Intelligence**

Awards & Honors

2026	**Third Place, ASME CIE Student Hackathon** ASME IDETC/CIE 2026
2024	**Best Graduate Presentation Award at NAPS 2024**
2023 - Present	**Cockrell School of Engineering Fellowship**
2023	**Mohamad Ali Safieddine Award for Academic Excellence** for ranking first across the AUB Maroun Semaan Faculty of Engineering and Architecture
2023	**ECE Distinguished Graduate Award** for ranking first among ECE graduates
2023	**Exceptional ECE Final Year Project Award** Power and Energy Systems
2021	**Best Pitch Award EPFL Tech4Impact Summer School**
2019 - 2023	**Dean’s Honor List** AUB Maroun Faculty of Engineering and Architecture - every given semester